

Szegedy Quantum Walk and Qubitization

Block encoding

Goal:

- ▶ We want to implement a (possibly non-unitary) operator A
- ▶ Quantum circuits can only implement unitary operators

Idea:

Embed A into a larger unitary U

State Embedding

We embed vectors into a larger Hilbert space using an ancilla qubit.

$$|0\rangle \otimes |b\rangle \longleftrightarrow \begin{pmatrix} b \\ 0 \end{pmatrix}, \quad |1\rangle \otimes |b\rangle \longleftrightarrow \begin{pmatrix} 0 \\ b \end{pmatrix}$$

Interpretation:

- ▶ The ancilla selects a block of the larger vector
- ▶ We are embedding $\mathbb{C}^n \rightarrow \mathbb{C}^{2n}$

Block Matrix Form

A unitary U acting on the larger space can be written as:

$$U = \begin{pmatrix} A & * \\ * & * \end{pmatrix}$$

Meaning:

- ▶ The top-left block encodes A
- ▶ Other blocks ensure U is unitary

Block Encoding Definition

Formally, U is a block encoding of A if:

$$(\langle 0| \otimes I) U (|0\rangle \otimes I) = A$$

Equivalently:

$$A = \Pi U \Pi \quad \text{where} \quad \Pi = |0\rangle\langle 0| \otimes I$$

Interpretation:

- ▶ Prepare ancilla in $|0\rangle$
- ▶ Apply U
- ▶ Project back to $|0\rangle$

Action on States

Apply U to embedded state:

$$U(|0\rangle \otimes |b\rangle) = |0\rangle \otimes A|b\rangle + |1\rangle \otimes |\text{garbage}\rangle$$

Key idea:

- ▶ Desired output appears in the $|0\rangle$ branch
- ▶ Other components ensure unitarity

Summary

- ▶ Block encoding embeds A into a unitary U
- ▶ Extraction via projection:

$$A = (\langle 0| \otimes I) U (|0\rangle \otimes I)$$

- ▶ Enables:
 - ▶ Hamiltonian simulation
 - ▶ Quantum walks
 - ▶ Qubitization

Setup

Let P be a reversible Markov chain.

Define:

$$O_P |0^n\rangle |j\rangle = \sum_k \sqrt{P_{kj}} |k\rangle |j\rangle$$

Define:

$$U_D = O_P^\dagger \text{SWAP } O_P$$

Equation

Apply O_P and SWAP:

$$\begin{aligned} |0^n\rangle|j\rangle &\xrightarrow{O_P} \sum_k \sqrt{P_{kj}}|k\rangle|j\rangle \\ &\xrightarrow{\text{SWAP}} \sum_k \sqrt{P_{kj}}|j\rangle|k\rangle \end{aligned}$$

Equation

Similarly:

$$|0^n\rangle|i\rangle \xrightarrow{O_P} \sum_{k'} \sqrt{P_{k'i}} |k'\rangle|i\rangle$$

Equation

Compute:

$$\begin{aligned} & \langle 0^n | \langle i | U_D | 0^n \rangle | j \rangle \\ &= \sum_{k, k'} \sqrt{P_{k'i} P_{kj}} \langle k' | j \rangle \langle i | k \rangle \\ &= \sqrt{P_{ji} P_{ij}} = D_{ij} \\ \Rightarrow & (\langle 0^n | \otimes I) U_D (| 0^n \rangle \otimes I) = D \end{aligned}$$

Block Encoding Interpretation

Thus:

$$U_D = O_P^\dagger \text{SWAP} O_P$$

is a block encoding of D .

- ▶ D is not unitary
- ▶ U_D is unitary

Qubitization

Define:

$$\Pi = |0^n\rangle\langle 0^n| \otimes I$$

$$W = U_D(2\Pi - I)$$

This is Szegedy's quantum walk operator.

States

Define:

$$|\psi\rangle = \sum_{i,j} \sqrt{\pi_j P_{ij}} |i\rangle |j\rangle$$

$$|\phi\rangle = \sum_{i,j} \sqrt{\pi_j P_{ij}} |j\rangle |i\rangle$$

Reflections

$$R_\psi = 2|\psi\rangle\langle\psi| - I$$

$$R_\phi = 2|\phi\rangle\langle\phi| - I$$

$$W = R_\phi R_\psi$$

Inner Product Structure

Compute:

$$\langle \psi | \phi \rangle = \sum_{i,j} \pi_j P_{ij} = 1$$

More generally, in eigenbasis:

$$\langle \psi_j | \phi_j \rangle = \lambda_j$$

Two-Dimensional Subspace

Define:

$$|\psi_j^\perp\rangle = \frac{|\phi_j\rangle - \lambda_j|\psi_j\rangle}{\sqrt{1 - \lambda_j^2}}$$

Then:

$$|\phi_j\rangle = \lambda_j|\psi_j\rangle + \sqrt{1 - \lambda_j^2}|\psi_j^\perp\rangle$$

Action of Reflections

$$R_\psi|\psi_j\rangle = |\psi_j\rangle \quad R_\psi|\psi_j^\perp\rangle = -|\psi_j^\perp\rangle$$

$$R_\phi|\phi_j\rangle = |\phi_j\rangle$$

Derivation of Rotation

Compute:

$$\begin{aligned} W|\psi_j\rangle &= R_\phi|\psi_j\rangle = |\phi_j\rangle \\ &= \lambda_j|\psi_j\rangle + \sqrt{1 - \lambda_j^2}|\psi_j^\perp\rangle \end{aligned}$$

Similarly:

$$W|\psi_j^\perp\rangle = -\sqrt{1 - \lambda_j^2}|\psi_j\rangle - \lambda_j|\psi_j^\perp\rangle$$

Equations

Thus in basis $\{|\psi_j\rangle, |\psi_j^\perp\rangle\}$:

$$W = \begin{pmatrix} \lambda_j & \sqrt{1 - \lambda_j^2} \\ -\sqrt{1 - \lambda_j^2} & \lambda_j \end{pmatrix}$$

Let:

$$\lambda_j = \cos \theta_j$$

Then eigenvalues:

$$e^{\pm i\theta_j}$$

Spectral Gap Transformation

Classical:

$$\lambda_2 = 1 - \gamma$$

Quantum:

$$\theta_2 = \arccos(\lambda_2) \approx \sqrt{2\gamma}$$

(Expand $\cos \theta \approx 1 - \frac{\theta^2}{2}$)

Quadratic Speedup

Classical convergence:

$$(1 - \gamma)^k \Rightarrow k = O(1/\gamma)$$

Quantum:

$$\theta \sim \sqrt{\gamma} \Rightarrow k = O(1/\sqrt{\gamma})$$

Quadratic speedup

From Phase Estimation to the Stationary State

Let W be the Szegedy walk with eigenpairs

$$W|\psi_j\rangle = e^{\pm i\theta_j}|\psi_j\rangle, \quad \cos \theta_j = \lambda_j,$$

and $\theta_1 = 0$ for the stationary component.

Given an initial state

$$|\Psi_{\text{init}}\rangle = \sum_j \alpha_j |\psi_j\rangle,$$

apply phase estimation for W :

$$\sum_j \alpha_j |\psi_j\rangle |\tilde{\theta}_j\rangle.$$

Project onto phase 0:

- ▶ Measure the phase register
- ▶ Post-select on outcome ≈ 0

$$\Rightarrow |\psi_1\rangle \quad (\text{coherent stationary state})$$

Sampling from the Stationary Distribution

The stationary eigenstate has the form

$$|\psi_1\rangle = O_P(|0^n\rangle \otimes |\sqrt{\pi}\rangle) = \sum_{i,j} \sqrt{\pi_j P_{ij}} |i\rangle |j\rangle.$$

Sampling:

- ▶ Measure the second register

$$\Pr[j] = \sum_i \pi_j P_{ij} = \pi_j.$$

Remarks:

- ▶ Success probability = $|\alpha_1|^2$ (overlap with $|\psi_1\rangle$)
- ▶ Use amplitude amplification to boost
- ▶ Total cost:

$$O\left(\frac{1}{\sqrt{\gamma}}\right) \quad (\text{from phase gap } \theta_2 \sim \sqrt{\gamma})$$

Key Insight

- ▶ Block encoding gives access to D
- ▶ Qubitization turns eigenvalues into phases
- ▶ Walk acts as rotation in 2D subspaces
- ▶ Phase estimation extracts eigenvalues faster

Conclusion

- ▶ Szegedy walk = qubitization of discriminant matrix
- ▶ Spectrum mapped: $\lambda \rightarrow e^{\pm i \arccos \lambda}$
- ▶ Leads to quadratic improvement in spectral gap